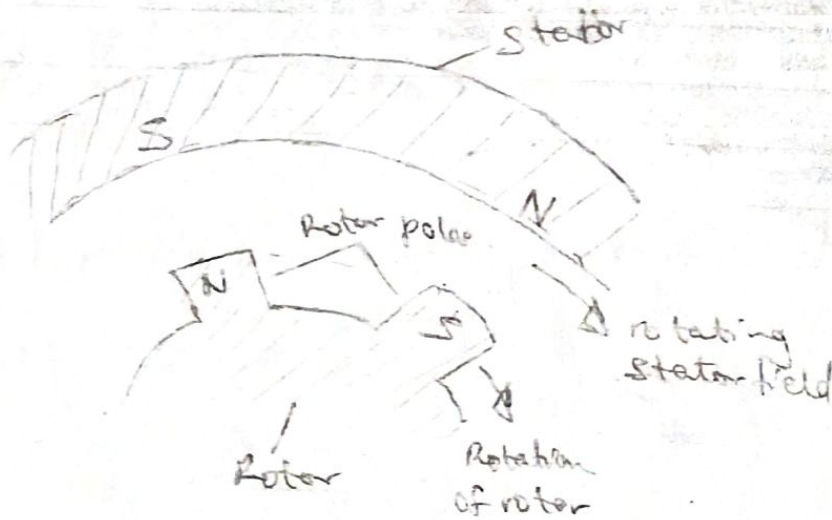


SYNCHRONOUS MACHINES

①

PRINCIPLE OF OPERATION OF A 3-PHASE SYNCHRONOUS MOTOR.

For the machine to operate as a motor the three-phase windings are connected to a 3-phase supply. This produces a rotating magnetic field in which there ^{are} no physical poles but the magnetic flux ~~the~~ rotate round the stator at synchronous speed. In the diagram below, the stator poles ~~are~~ (not physical) are indicated by the letters N and S.



The figure shows the torque produced by a synchronous motor.

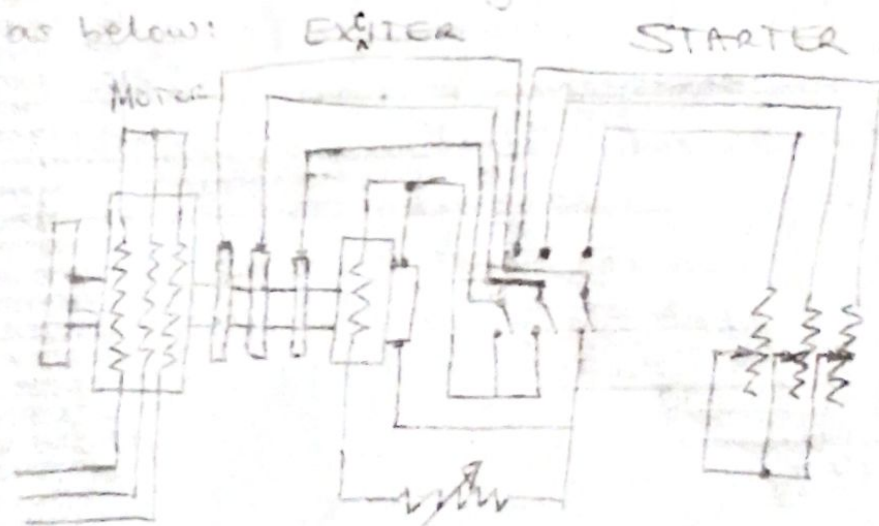
Considering the stator field as moving clockwise and its effect on the salient rotor, the letter N and S on the stator indicates the position of the rotating stator poles at one particular instant.

Suppose the stator is stationary. The stator field will be rotating at synchronous speed, such that, each rotor pole will be subjected to both an attractive and a repelling force every

cycle of the ~~ap~~ supply frequency as it is passed by two stator poles of opposite polarities. The inertia of the rotor will not allow it to move in either direction, and therefore when the rotor is stationary the resultant torque is zero. Thus ~~the~~ a synchronous motor is not self-starting.

If the rotor is made to rotate at or near the same speed as the stator field (i.e. synchronous speed) the stator and rotor poles of opposite polarity will then "lock" together, producing a torque which will cause the rotor to continue rotating at synchronous speed. If the motor slows down it will fall out of step (i.e. out of synchronism, with the stator), the torque will be reduced to zero, and the rotor will stop. Thus the synchronous motor can run only at synchronous speed.

To achieve this operation the ~~ac~~ stator winding and the d.c. field winding (on a salient pole rotor) are arranged as below:



The figure shows a,

THREE-PHASE SYNCHRONOUS MOTOR WITH A SIMPLIFIED STARTING ARRANGEMENT.

NR In itself, the synchronous motor has no starting torque

and special starting arrangement are necessary. For this purpose the a.c. voltage is applied to the armature and a separate d.c. supply, usually 110V is connected to the rotor through sliprings.

STARTING A SYNCHRONOUS MOTOR.

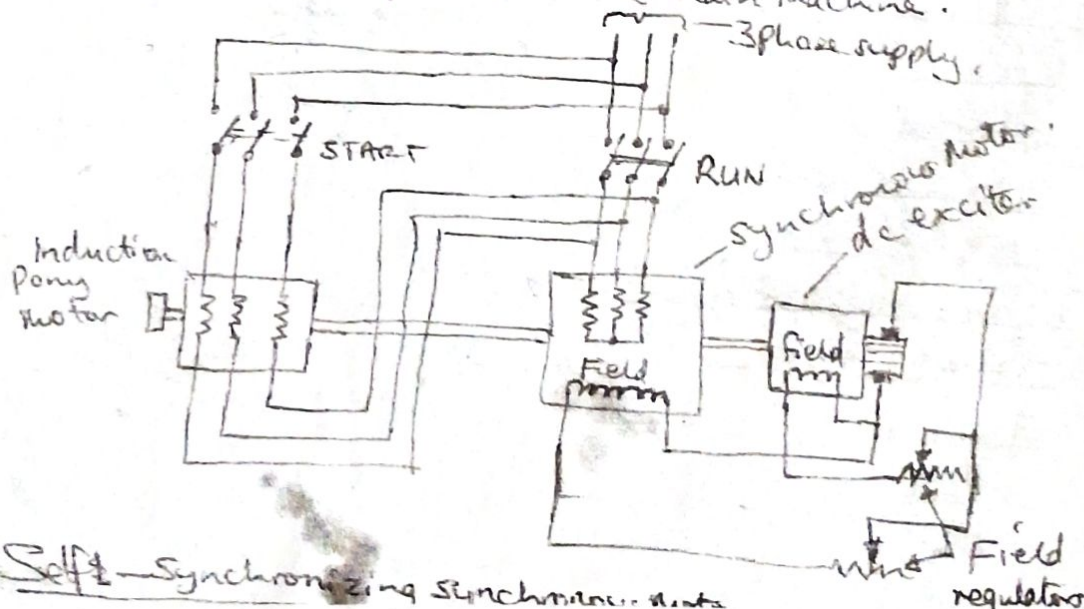
- Since the machine is not self starting, a provision is required to run the machine up to synchronous speed before switching in the supply.

Two starting methods are employed:

- pony motor &
- induction starting

(i) Pony motor.

This is a small motor (usually coupled an induction motor), coupled to the synchronous motor shaft. This motor drives the main motor up to or near synchronous speed on no-load. The excitation is then switched on and the machine synchronized with the 3-phase supply. The induction motor can then be switched off and the load applied to the main machine.



Self-Synchronizing synchronous motor

Owing to slip, the induction motor will drive the synchronous machine at a speed a little less than synchronous speed; but, unless it is a large machine, the main motor will be pulled into synchronism when the supply is switched on. However, to enable synchronous speed to be reached, the induction motor may have two poles less than the synchronous motor so that its synchronous speed will be higher than that of the machine. This will take the main motor right through its own synchronous speed and allows it to be synchronised at its correct speed.

A development of this method is to connect the stator windings of the induction motor and ~~the~~ the synchronous motor in series. This method makes the machine self-synchronizing as per the diagram already drawn.

When the 'Start' switch is closed, the induction motor drives the main machine and passes its starting current through the stator windings of the synchronous motor, producing a weak rotating field. As synchronous speed is reached, this rotating field will be strong enough to lock the rotor, whose excitation will also have become stronger with increase of speed. Thus, ^{the} motor synchronizes itself and the rotor continues to rotate at synchronous speed. The 'run' switch can now be closed, the start switch opened and load applied to the motor. This method allows light-load starting.

(ii) ~~the~~ Induction Starting

In this case, the motor is designed so that it starts as an induction motor which allows starting ~~on~~ on light loads. The rotor poles carry a cage-type winding called a 'damper winding'. With the excitation

off and the rotor windings short circuited, the motor is started like an induction motor with direct-on-line starting for small machines, and auto-transformer or star-delta starting for larger machines. When maximum speed has been attained which ~~is~~ will be at a little under synchronous speed, the normal dc excitation ~~excitation~~ is switched on and the motor will be pulled into synchronism. This means ~~that~~, a sudden increase in speed to synchronous speed, which causes a heavy current to be drawn from the supply for a short time. Starters must be capable of withstanding this current surge.

Speed Characteristics

- Constant Speed.
- Always runs at synchronous speed, ~~but slight~~ ^{adjusts} reduction may occur on overload.
- No Speed Control

Applications

- power-factor improvement owing to its ability to run at a leading power factor.
- Suitable for continuous service to avoid unnecessary stalling, i.e. used for driving large ventilating fans, compressors, pumps, motor-generator sets etc.

Reversal of direction of rotation

By changing over the connections of two phases (any two phases) the direction of rotation will be reversed.

⑥ 5

IEE REGULATIONS REGARDING ROTATING MACHINERY (AC & DC) (as per the 14th edition)

deals with the control of motors:

Reg A68-65^A Every motor shall be provided with a means of starting and stopping and this control should be placed within the reach of the operator.

In addition motors exceeding 0.37 kW should be provided with a control gear comprising the following devices:

- (a) means of preventing self-starting in the event of restoration of supply after failure (due to stoppage or drop in voltage)
- (b) protection means of protection against over current in motor and its associated circuits (cables).
- (c) means of isolating the motor and all its associated apparatus.

Reg A66 states that the cables carrying the starting, accelerating, and running current of the motor shall be of ratings of at least equal to the full load current rating of the motor.

Reg A67 to 68:

— requires that a final sub-circuit supplying a motor shall be protected by a fuse or a circuit breaker of current rating not greater than twice the cable rating.

Reg A69: ~~The~~

The size of the cables for the rotor circuit of slipring or commutator induction motors must be suitable for starting and running

Reg B23: - the volt drop between the consumer's terminals to motor terminals should not exceed 2.5 percent of the declared or nominal voltage.

Reg C8-10 and 14:

Where there is danger of fire or explosion, the motor and its control gear shall be of the flame proof construction. Also in dust-laden situation the motor enclosure shall be of ~~the~~ a type to exclude dust.

~~Reg C14~~ - In situations where the temperature of the surrounding atmosphere may cause over-heating, the motor shall be down rated, or the insulation used shall be of special material. Means of forced ventilation or pipe-ventilated enclosures, can be used as an alternative.

Reg F14

- states that all live parts of open type resistor and machine control gear must be guarded to prevent their being inadvertently touched while in service.

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